How-To Library

Informatica

Set up OAuth with Okta

© Copyright Informatica LLC 2024. Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.htm.

Abstract

If your organization is licensed and configured for SAML, you can use OAuth to start a REST API session using JSON web token (JWT) access tokens. This article explains how to configure OAuth using Okta as the identity provider.

Supported Versions

- Informatica Intelligent Cloud Services May 2024

Table of Contents

Overview. 2
Step 1. Register an OIDC native application. 3
Step 2. Add an Okta authorization server. 5
Step 3. Register the identity provider in Informatica Intelligent Cloud Services. 10
Fetch JWT tokens. 10

Overview

If you are an Informatica Intelligent Cloud Services administrator and your organization is licensed and configured to use SAML, you can set up OAuth using Okta as the identity provider so that users can start REST API sessions using JSON Web Token (JWT) access tokens.

Using JWT access tokens to log in to Informatica Intelligent Cloud Services is similar to using SAML assertions. However, unlike using SAML assertions, users must retrieve the JWT access tokens to use in login requests.

Note: To set up OAuth, at least one native user account with administrative privileges must exist in the Informatica Intelligent Cloud Services organization. This user account can be removed after the OAuth setup is complete.

Before your Informatica Intelligent Cloud Services organization can use JWT access tokens, complete the following tasks:

1. Configure the organization to use SAML for authentication, and set up users as SAML users.
   For more information, see Configure SAML SSO using Okta in IICS. If you want to use user provisioning through SCIM, also see Setting up SCIM with Okta.
2. Register an OpenID Connect (OIDC) native application in Okta.
3. Add an Okta authorization server.
4. Register the identity provider using the IdentityProviders resource in the Informatica Intelligent Cloud Services REST API.

To log in, users obtain a JWT access token from the identity provider and include the token in a loginOAuth POST request. The token can be used for one REST API session. If the login request is successful, the response includes a session ID to use in subsequent API calls.

This article describes how to register an OIDC native application and add an Okta authorization server. For more information about using Okta, see the Okta documentation.

For information about SAML configuration, see SAML single sign-on.

For information about registering an identity provider and logging in to Informatica Intelligent Cloud Services using JWT access tokens, see the following topics in *REST API Reference*:

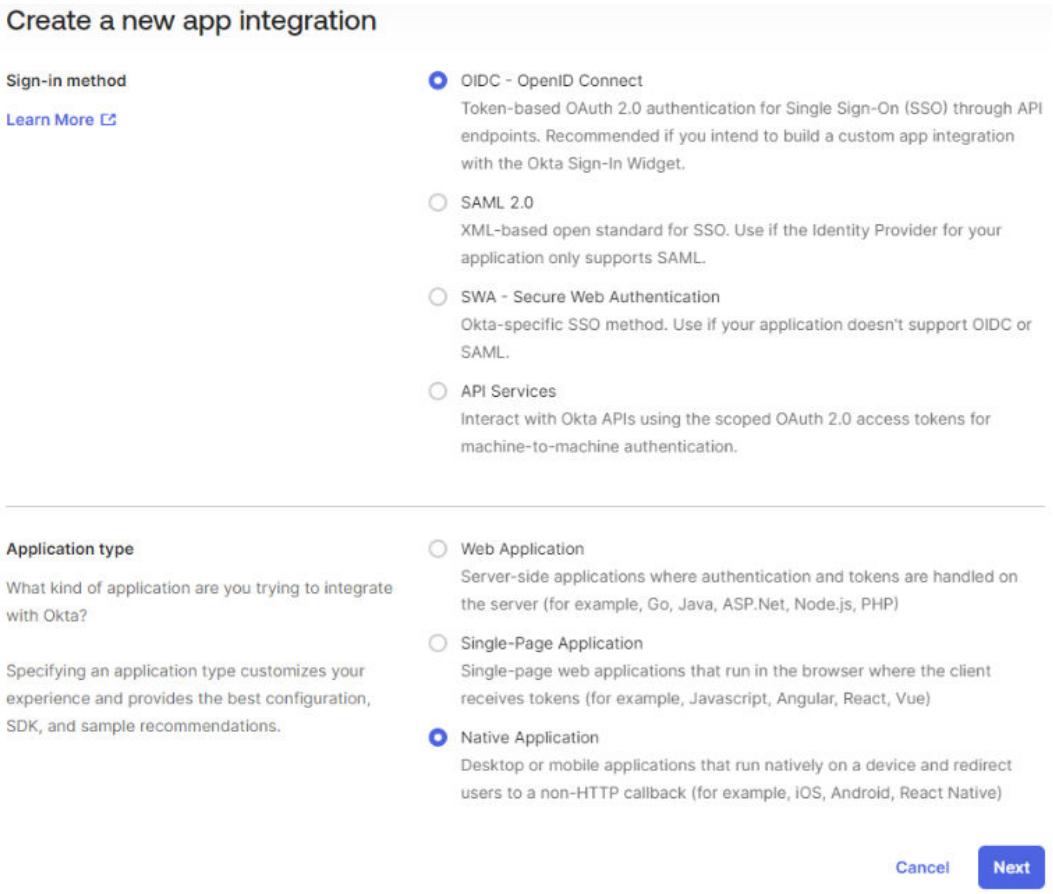
- Identity providers

- loginOAuth

Step 1. Register an OIDC native application

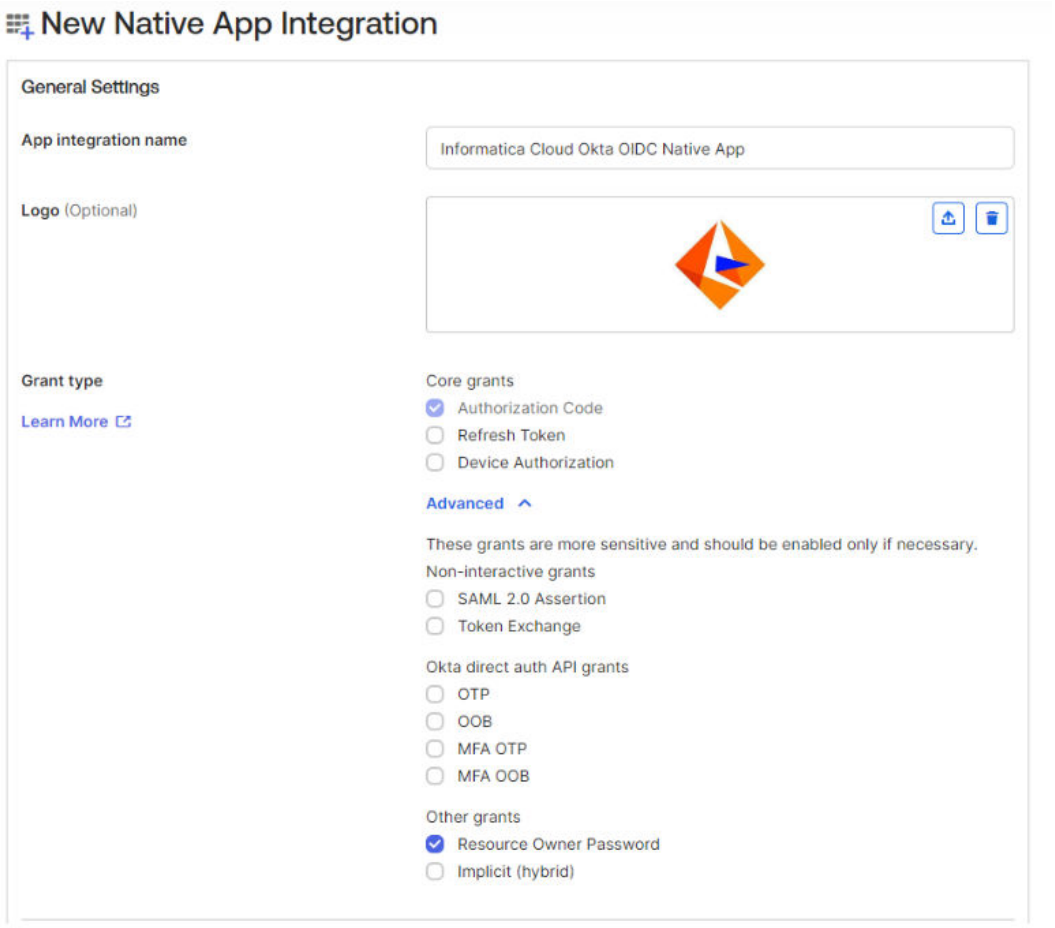
Register a client application to use for Oauth on the **App registrations** page.

1. Log in to the Okta portal as an Okta administrator.
2. Select **Applications** > **Applications** and click **Create App Integration**.

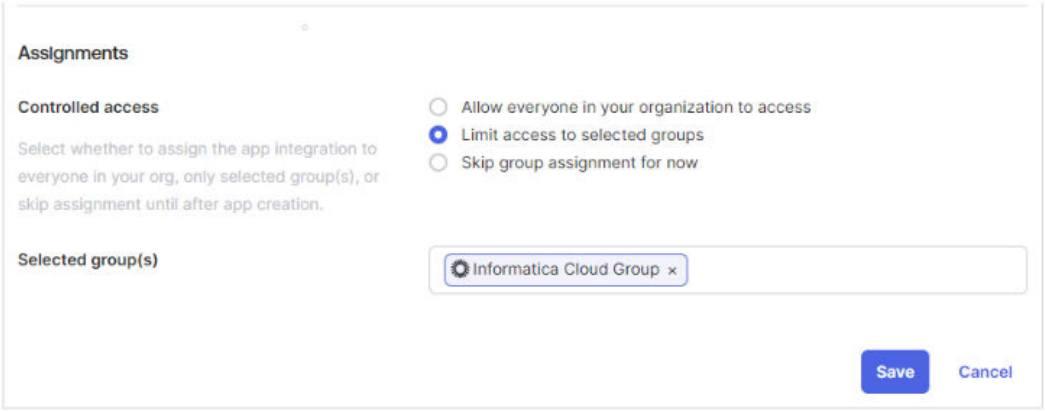


3. For sign-in method, select **OIDC - OpenID Connect**.
4. For the application type, select **Native Application**.
5. Click **Next**.

The **New Native App Integration** page appears.



6. For the application integration name, enter a display name.
7. Optionally, upload a logo image.
8. For the grant type, ensure that **Authorization Code** is selected.
9. Expand **Advanced** and under **Other grants**, select **Resource Owner Password**.
10. Scroll down to **Assignments**.

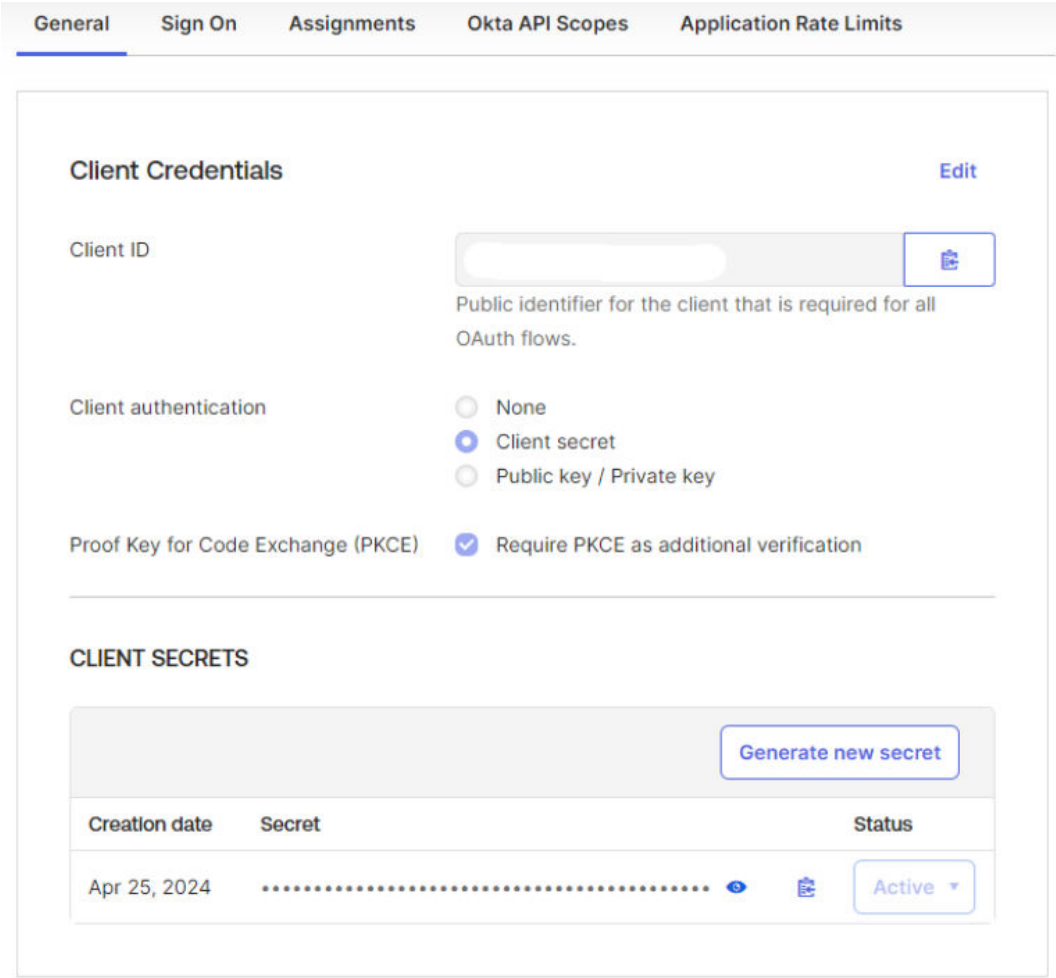


11. In the **Controlled access** section, you can allow everyone in the organization access to the application or limit access to certain groups. To limit access to certain groups, start typing the names of each group that you want to allow access to select them from the list.

    Optionally, you can assign groups and assign people later. To do so, select **Skip group assignment for now**.

    Note: If you want to allow access to everyone, review your security posture carefully before you enable the option.
12. Click **Save**.
13. On the **General** tab under **Client Credentials**, select **Client secret** and **Require PKCE as additional verification**.

    This action generates a client secret. If a client secret doesn't appear, click **Generate new secret**.



14. On the **Assignments** tab, assign people or groups.

    If you assigned groups in step 11 and you don't want to assign people, you can skip this step.

Step 2. Add an Okta authorization server

Add an Okta authorization server for the application.

1. Log in to the Okta portal as an Okta administrator.
2. Select **Security** > **API**.

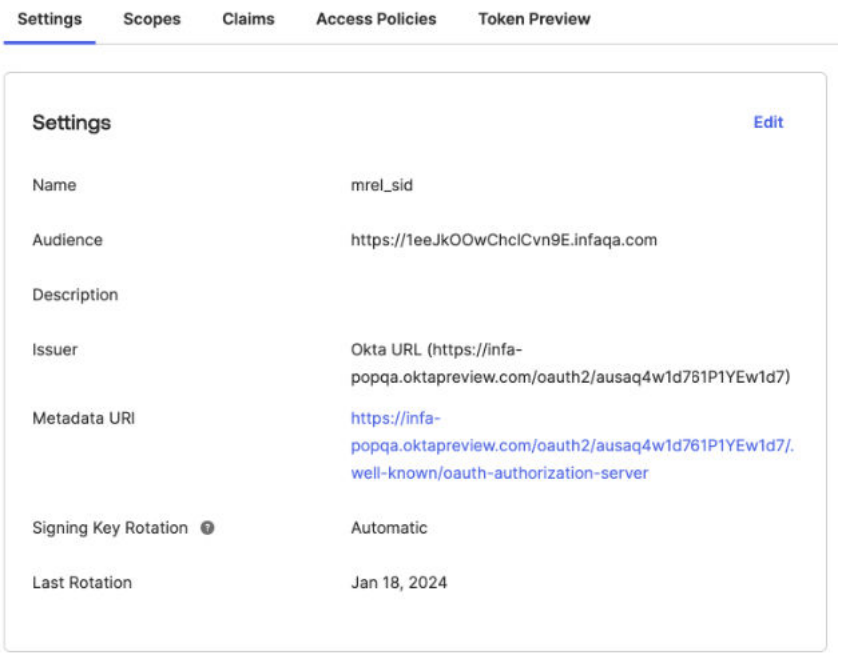
On the **Authorization Servers** tab, click **Add Authorization Server**.

3. Enter a display name.
4. For the audience, enter the SAML entity ID that's configured in the organization's SAML setup. The JWT access token's audience claim maps to the SAML entity ID.

   The audience claim can be one of the following values:

   - `api://<orgId>.<cloud provider>-<region>.informaticacloud.com`
   - `api://<cloud provider>-<region>.informaticacloud.com.<orgId>`
   - `api://<cloud provider>-<region>.informaticacloud.com/<orgId>`
   - `https://<orgId>.<cloud provider>-<region>.informaticacloud.com`
   - `https://<cloud provider>-<region>.informaticacloud.com.<orgId>`
   - `https://<cloud provider>-<region>.informaticacloud.com/<orgId>`

   You can find the SAML entity ID in the service provider metadata XML file. The SAML entity ID is the value of the entity ID attribute in the entityDescriptor object. You can download the service provider metadata XML file from the **SAML Setup** page in Administrator. For more information, see Downloading the service provider metadata.
5. Provide a description and click **Save**.
6. On the **Settings** tab, copy the **Metadata URI** value.



7. Paste the metadata URI into a browser URL.

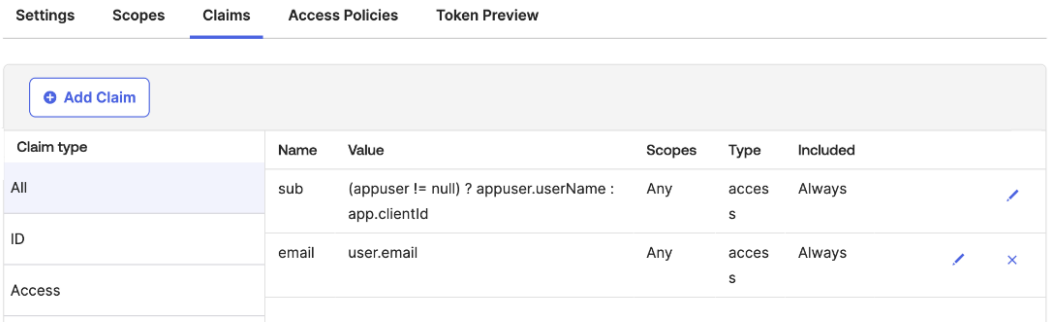
The browser returns JSON output with Okta authorization server values. Note the values for the following fields, which you'll need when you register the identity provider in Informatica Intelligent Cloud Services:

Field	Description
issuer	Okta authorization server issue URI.
jwks_uri	Okta authorization server JSON web key sets URI.
token_endpoint	Okta authorization server access token endpoint.

Tip: Copy the output to a text editor that allows formatting so you can easily find the field values that you need.

8. On the **Claims** tab, a default subject claim named "sub" should already exist. The value of the sub claim should map to the SAML user name.

   If the value maps to the SAML user name, no further action is required.

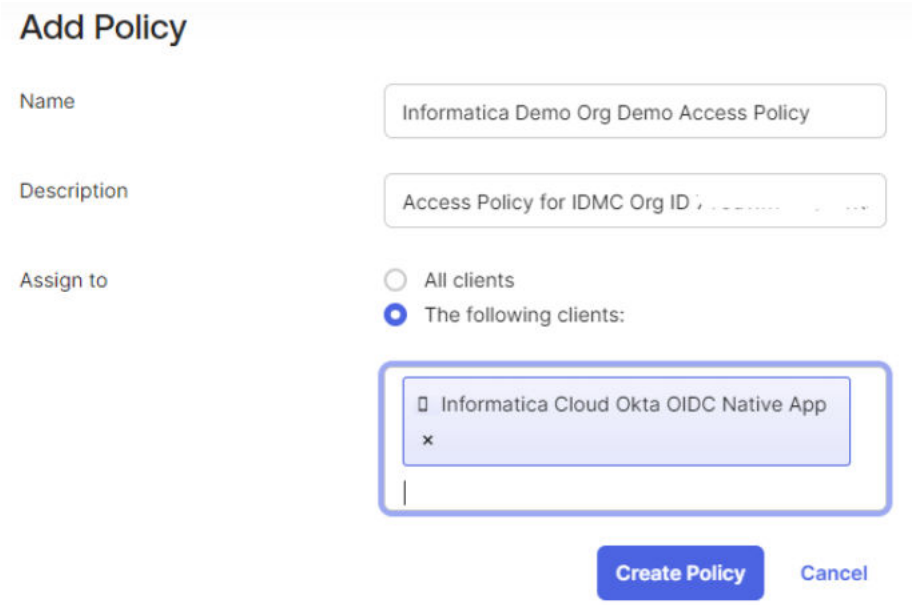


9. If the SAML user name value isn't available in the sub claim, you can edit the sub claim value to include the Okta attribute that represents the SAML user name, or you can create a new claim.

   To create a new claim, click **Add Claim**. Configure the following fields:

Field	Description
Name	Enter a name for the claim. The name must map to the SAML user name.
Include in token type	Select **Access Token**.
Value type	Select **Expression**.
Value	Enter the Okta attribute value that maps to the SAML user name. Use Okta Expression Language syntax.
Disable claim	Ensure that this option is not enabled.
Include in	To restrict the claim to specific scopes, select **The following scopes** and select the relevant scopes from the list. You can select multiple scopes. By default, **Any Scope** is enabled.

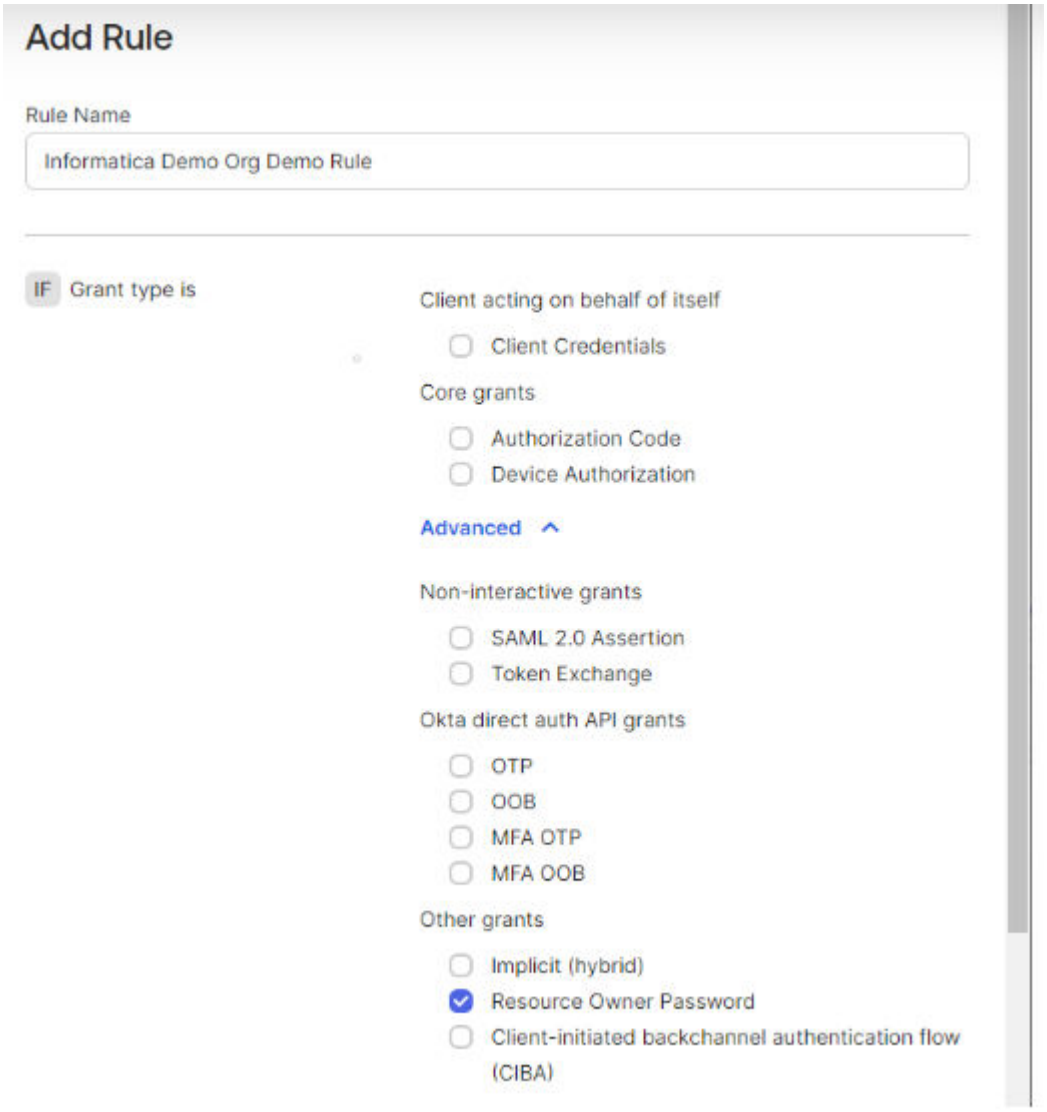
10. On the **Access Policies** tab, click **Add Policy**.



11. Provide a display name and a description.
12. In **Assign to**, select **The following clients** and start typing the name of the OIDC native application you created earlier to select it from the list.

    You can select multiple clients.
13. Click **Create Policy**.

14. On the **Access Policies** tab, click **Add rule** for the access policy you created in the previous step.



15. Enter the rule name and configure the following options:

Option	Description
IF Grant type is	Clear all checkboxes and then for **Other grants**, select **Resource Owner Password**.
AND User is	Select any user assigned to the application, or to restrict access further, select users assigned to the application who are members of any of the groups that you specify.
AND Scopes requested	Select **The following scopes** and then select **openid** from the list.
AND Access token lifetime is	Accept the default value or revise if necessary.

16. Click **Create rule**.
17. Optionally, you can verify the configuration on the **Token Preview** tab.

Step 3. Register the identity provider in Informatica Intelligent Cloud Services

Register the identity provider using the Informatica Intelligent Cloud Services REST API.

Before you register the identity provider, be sure you have the following information:

- A native, non-SAML Informatica Intelligent Cloud Services user account's user name and password. If you don't have access to a native, non-SAML user account, create one temporarily.
- The Informatica Intelligent Cloud Services organization ID.
- The issuer URI and JSON web key sets URI of the Okta authorization server that was provided in the JSON response in step 7 of "Step 2. Add an Okta authorization server" on page 5.
- The name of the custom access claim, if applicable and created earlier.

To register the identity provider, log in to Informatica Intelligent Cloud Services using the native, non-SAML user credentials, and then send a request using the IdentityProviders API resource.

For information about using the IdentityProviders API resource, see loginOAuth.

Fetch JWT tokens

Before you can log in to Informatica Intelligent Cloud Services using the loginOAuth resource, you need to fetch a JWT access token from Okta. Include the value of the JWT access token in the oauthToken field in the loginOAuth request.

Send a POST request using the token endpoint that was returned in the JSON response in step 7 of "Step 2. Add an Okta authorization server" on page 5.

For the header, set the content-type to `application/x-www-form-urlencoded`. Use basic authorization with the following information:

Field	Description
username	The Okta OIDC native application client ID.
password	The Okta OIDC native application client secret.

Use the body type `x-www-form-urlencoded` and include the following keys:

Key	Description
grant_type	Use the following value: password
scope	Use the following value: openid
username	User name registered in the Okta identity provider.
password	Password for the user registered in the Okta identity provider.

A request might look similar to the following example:

```
POST <Okta authorization server access token endpoint API path> HTTP/1.1
Host: <Okta authorization server access token base>
Accept: application/json
Content-Type: application/x-www-form-urlencoded
Authorization: Basic <Base64Encode(<Okta OIDC native application client ID>:<Okta OIDC native
```

```
application client cecret)>
grant_type=password&username=<human/machine or service user registered in Okta
IdP>&password=<Password of human/machine or service user registered in Okta IdP>&scope=openid
```

If you use curl, the request might look similar to the following example:

```
curl --location '<Okta authorization server access token endpoin>' \
--header 'Accept: application/json' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Basic <Base64Encode(<Okta OIDC Native Application Client ID>:<Okta
OIDC native application client secret)>' \
--data-urlencode 'grant_type=password' \
--data-urlencode 'username=<human/machine or service user registered in Okta IdP>' \
--data-urlencode 'password=<Password of human/machine or service user registered in Okta IdP>' \
--data-urlencode 'scope=openid'
```

A successful response includes the JWT access token in the access_token field. Copy the token and include it in a loginOAuth POST request to log in to Informatica Intelligent Cloud Services and start an API session.

For information about using the loginOAuth API resource, see loginOAuth.

Author

Informatica Cloud Trust Team